

Donald Moratz

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Research Interests

I study how information environments shape political behavior, with a focus on state media, foreign influence, and civic space in authoritarian and developing-country contexts. My work combines causal inference with text-as-data methods, including transformer models and LLMs applied to large multilingual news corpora.

Education

- Ph.D. **University of Pennsylvania**, *Political Science*, International Relations, Methods, and Comparative Politics, Expected 2027
- M.A. **University of Pennsylvania**, *Statistics*, Concurrent with Ph.D., Expected 2027
- M.A. **Duke University**, *Economics and Political Science*, May 2019
- M.A. **University of St. Andrews**, *Economics and International Relations*, June 2013

Awards & Honors

- Penn Grad Talks, Social Science Winner, 2026 (*competitive university-wide research communication competition*)
- Steven B. Feirson Graduate Fellow, 2023-2024
- Forever Duke Student Leadership Award, 2019
- Duke Economics Student Leadership Award, 2019

Working Papers

- “The Signaling Gap: State-Owned Media and the Strategic Management of Sentiment during the Russian Invasion of Ukraine” with Sarah Gerstein. 2025.
LLM-based sentiment analysis of state-owned media in Russia’s near abroad shows that the 2022 invasion shifted coverage of Russia more negatively in security-allied states than in neutral ones.
- “Separating Rate from Composition: A Zero-Loss Decomposition of Electoral Change”. 2025.
Introduces a zero-loss decomposition (RCVD) of electoral change into rate, composition, and volume effects; applied to recent U.S. presidential elections, shows rightward Hispanic shifts in 2024 fully neutralized their Democratic-leaning demographic growth.

- “Tracking Civic Space in Developing Countries with a High-Quality Corpus of Domestic Media and Transformer Models” with Jeremy Springman, Erik Wibbels, Fatih Serkant Adiguzel, Mateo Villamizar Chaparro, Zung-Ru Lin, Diego Romero, Mahda Soltani, Hanling Su, and Jitender Swami. 2025.
New monthly event data on civic space activity across 65 developing countries (2012-2024), built from a transformer-based pipeline over 120M+ articles from 350+ domestic media outlets.
- “Foreign Influence by Authoritarian Governments: Introducing New Data and Evidence” with Jeremy Springman, Fatih Serkant Adiguzel, Mateo Villamizar Chaparro, Zung-Ru Lin, Diego Romero and Erik Wibbels. 2025.
LLM-based analysis of a 120M-article corpus across 66 aid-receiving countries (2012-2025) detects a previously undetected surge in Russian diplomatic and hard-power activity before the Ukraine invasion.
- “#DataForUkraine: Adapting Social Science Tools for Crisis Response” with Ernesto Calvo, Zung-Ru Lin, Olga Olnuch, Graeme Robertson, Jeremy Springman and Erik Wibbels. 2022.
A near real-time, social media-based event detection system built for policymakers during the 2022 invasion of Ukraine, with reflections on crisis-response data infrastructure.

Policy Reports

- “Violence Against Journalists and Reporting on Civic Space”, with Erik Wibbels 2023
- “Technical Report on the Production of Civil Space and RAI Event Count Data” with Jeremy Springman and Erik Wibbels, 2021
- “Legal Changes and Protest: Evidence from High-Frequency Data” with Jeremy Springman and Erik Wibbels, 2021
- “What Predicts Changes in Civic Space? Evidence from Forecasts in Serbia” with Jeremy Springman and Erik Wibbels, 2021

Research Experience

Research Manager, *DEVLAB@Duke Machine Learning for Peace*, Durham, NC, 2020-2022

- Promoted to Research Manager from Research Project Supervisor; managed a research team of approximately 10 (postdocs, PhD students, data scientists, and RAs) producing data infrastructure and forecasting outputs for the USAID-funded INSPIRES Consortium on civic space resilience.
- Led training data construction, codebook development, and validation for transformer-based classification pipelines translating multilingual news articles and coding them into civic space and foreign influence event categories; supervised pipeline engineering and model development in Python.
- Co-authored four working papers using DEVLAB data and infrastructure, including lead authorship on the civic space methodology paper.
- Produced quarterly forecasts of civic space activity (protests, arrests, civil society restrictions) across 65 developing countries, delivered to USAID missions and INSPIRES Consortium partners; led annual policy report production.
- Built original public-facing Shiny dashboard for live civic space data, still maintained at web.sas.upenn.edu/mlp-devlab..
- Led transition of forecasting infrastructure to a GitHub-based workflow; drove analytical improvements in R; led hiring process for two data scientists.

Research Intern, Division for Social Policy and Development, *United Nations*, New York, NY, 2018

- Conducted data analysis in R on cross-national disability metrics across 10 developing countries; contributions used in the UN Flagship Report on Disability and Development.
- Supported organization of the 11th Conference of State Parties to the Convention on the Rights of Persons with Disabilities (CRPD), including research, briefings, and side-event documentation.

Conference Presentations

- 2026 *The Signaling Gap*. APSA Annual Meeting (accepted).
- 2026 *The Signaling Gap*. Society for Political Methodology (PolMeth) Annual Meeting (accepted).
- 2026 *The Signaling Gap*. NLP Tutorial, SICSS-Penn (invited).
- 2025 *Separating Rate from Composition*. PolMeth Annual Meeting.
- 2025 *Foreign Influence by Authoritarian Governments*. NPSA.
- 2025 *Separating Rate from Composition*. ESOC.

Workshop Presentations

- 2026 *The Signaling Gap*. UPenn International Relations Workshop.
- 2026 *The Signaling Gap*. UPenn PDRI Workshop.
- 2026 *The Signaling Gap*. UPenn MEDIATED Workshop.
- 2025 *The Signaling Gap*. UPenn PDRI Workshop.
- 2025 *The Signaling Gap*. UPenn International Relations Workshop.
- 2025 *Tracking Civic Space*. UPenn PDRI Workshop.

2024 *Separating Rate from Composition*. UPenn PDRI Workshop.

Teaching

- STAT 5710: Modern Data Mining, Head TA 2026
- GAFL 5320: IDev Policy Research Shop, TA 2026
- Penn Political Science Ph.D. Math Camp, 2023, 2025
- PSCI 6800: Advanced Statistical Analysis, TA 2023, 2025
- Wharton Data Science Academy, Teaching Fellow, 2025
- PSCI 1408: War, Strategy and Politics, TA 2025
- PSCI 1800: Introduction to Data Science, TA 2024

Service

- Editorial Assistant, Political Analysis 2025-2026
- Reviewer, Political Analysis and Sociological Methods and Research
- Political Science Graduate Student Organization Co-Chair, 2023-2024, 2025-2026
- SASGov Student Representative, 2026
- Professional and Graduate Student Council Representative, Duke University, 2018-2019
- Master of Economics Student Council, Duke University, 2018-2019

Other Professional Experience

Financial Analyst II, *Tempur Sealy International*, Lexington, KY, 2013-2017

- Promoted from Financial Analyst I to Financial Analyst II
- Developed ad hoc models and analyses to support various functions within the business, providing support to sales, operations, marketing, accounting and other various finance teams using SQL and Oracle, including introducing new methods of analysis such as elasticity modeling
- Revamped forecast/budgeting processes to remove at minimum 20 hours of monthly work through improving both methods of analysis and applying greater Excel functionality reducing time spent on low value forecasting
- Finance lead on negotiations to launch Tempur branded products on Amazon.com, responsible for pricing, business cases and marketing plans

Research & Technical Skills

Methods	Causal inference, quasi-experimental design, sensitivity analysis, forecasting, spatial analysis, NLP and text-as-data, transformer models, LLM applications, sentiment analysis
Languages & Tools	R, Python, SQL, Git, LaTeX, Quarto
Selected R Libraries	tidyverse, data.table, fixest, estimatr, MatchIt, sensemakr, sf, ggplot2, patchwork, modelsummary

Selected pandas, numpy, scikit-learn, xgboost, seaborn, matplotlib
Python
Libraries